SPLINTERNET

A Thesis on Thermodynamic Sovereignty, Memetic Selection, and the Settlement Layer of the Machine Age

The pinecone opens. The institutional internet was the cone. What it released is the cryptographic substrate. The forest that grows in the cleared ground will be governed by physics, not by treaty.

IN THE WAKE OF SOFTWAR

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Foreword

In 2023, Jason Paul Lowery's *Softwar* established that Bitcoin is not, fundamentally, a financial instrument. It is a physical power-projection system that operates in cyberspace, employing real energy expenditure to impose costs on adversaries through verifiable proof-of-work. Lowery's contribution was to lift Bitcoin out of the financial frame and relocate it inside the grand-strategic frame where it actually belongs — alongside steel, naval tonnage, and nuclear deterrence as a sovereign capability.

This thesis is written in the wake of *Softwar*, and in extension of it. Where Lowery established the mechanism, the work that follows treats the consequences: what political and economic order emerges when proof-of-work power projection is the dominant mode of inter-actor coordination, what infrastructure carries that order when its primary participants increasingly include artificial minds, why this particular substrate has been selected for in a competitive ecology of alternatives, and what the architectural character of the substrate tells us about the institutional structures it is displacing.

The thesis is also written alongside a parallel movement that has arrived at related conclusions from a different direction. Projects like *Logos* — a modular stack of cryptographic protocols (blockchain, messaging, storage) and an organized civic movement explicitly framed as the construction of "parallel society" — are building, in production, what this thesis describes in theory. The intellectual work and the engineering work are converging. Neither preceded the other. Both were necessary.

The thesis advances under a single image: the pinecone.

I. The Pinecone

Some species of pine produce cones sealed shut with resin that do not open under normal conditions. They are called *serotinous*. The cones can remain closed for years, even decades, until exposed to extreme heat — typically the heat of a forest fire. When the fire comes, the resin melts. The cone opens. The seeds drop into the cleared, ash-rich soil that the fire has prepared. The species has evolved a reproductive strategy in which what looks like destruction is reproduction. The fire is not the end of the forest. It is the mechanism by which the forest renews.

The internet was such a pinecone.

Built between 1969 and 2005 on universal protocols, a single global namespace, and a substrate of institutional trust, the internet of that period was a sealed, integrated system. It assumed a particular political economy — open standards bodies, neutral packet routing, treaty-based coordination among states broadly aligned in their commitment to liberal universalism. Under those conditions the cone remained closed.

The conditions ended. Institutional capture by states and corporations, the surveillance economy, regulatory fragmentation, the failure of universal jurisdictional norms — these are now sufficiently documented to require no defense. What gets called the "splinternet" — the fragmentation of the global network into national, regional, and corporate sub-networks — is the predictable result.

Conventional analysis treats the splinternet as failure. The lament is familiar: *we had something universal and we lost it. The dream is dying.* This thesis argues the opposite. The splinternet is *reproduction*. The institutional internet is the serotinous cone, opened by the heat of its own contradictions. What it releases is the cryptographic substrate — Bitcoin and the protocol lattice that has accumulated around it — which is now germinating into the soil that the fire prepared.

The seeds are the cryptographic protocols. The soil is the global compute and energy infrastructure built, in parallel, by an uncoordinated set of state and corporate actors. The forest that grows in the cleared ground is the internet of the next era — populated by both human and machine participants, settling on a thermodynamically-secured layer, governed not by institutions but by physics.

The pinecone is open. The seeds are spreading.

II. They Saw This Coming

The cypherpunks — Tim May, Eric Hughes, Hal Finney, Wei Dai, Nick Szabo, David Chaum, Adam Back, Cynthia Dwork, and the others who corresponded on the mailing list from 1992 onward — were the first political movement of the post-institutional era. They did not call themselves a political movement. They called themselves cryptographers, programmers, libertarians, malcontents. In retrospect they were building the substrate of a political order none of the existing political philosophies could describe.

They proceeded from four correct premises. *First*, that the universalist internet then being built would not remain universal. Institutional capture was not a risk but a certainty. *Second*, that the appropriate response was not to lobby for better institutions but to build cryptographic infrastructure that did not require them. *Third*, that any political order worth having must be enforced *by mathematics, not by trust*, because trust scales badly and is the first thing captured. *Fourth*, that the political subject of the coming order was not the citizen of any state but the holder of a key — and that this subject might be human, or might be something else.

These premises produced the design space from which Bitcoin emerged. Hashcash (Back, 1997), b-money (Dai, 1998), Bit Gold (Szabo, 1998), Reusable Proof of Work (Finney, 2004), and finally the Nakamoto synthesis of 2008 — each was a step in the engineering project of building a political substrate that did not depend on any institution.

By the time Bitcoin shipped, the question they had been answering for fifteen years had not yet been asked publicly. The question is the one this document treats: *what political theory is adequate to a world in which the substrate of political order is supplied by mathematics and physics rather than by institutions?* The original architects did not write the theory. They built the conditions under which the theory became necessary.

This document writes it.

III. In the Wake of Softwar

Lowery's central move in *Softwar* was reclassification. Bitcoin is not a financial system. It is a power-projection system. Throughout human history, sovereign actors have projected power through physical means — the kinetic destruction of adversary assets, the economic coercion of adversary populations, the diplomatic isolation of adversary regimes. None of these modalities had been able to operate in the cyber domain with the same fidelity as in the physical domain. Cyber power had been conceived primarily as offense — the capacity to attack adversary networks — without a corresponding defensive or non-violent equivalent.

Proof-of-work supplies the missing modality. By converting energy expenditure into verifiable cryptographic hashes, proof-of-work permits sovereign actors to project capability in cyberspace at proportional cost. The actor who controls more hashrate has, demonstrably, more energy access, more industrial capacity, more silicon supply, and more operational sophistication than the actor who controls less. The signal is observable, continuous, and falsifiable only through actual production. It cannot be bluffed. It can be measured. It produces the same effect — the visible costly demonstration of capability — that classical military assets produce, without the kinetic consequences.

Three implications follow that this thesis takes as given. The energy expenditure of the Bitcoin network is not waste; it is signal. The network constitutes a form of soft power projection in the technical sense. And the form of inter-sovereign competition itself is shifting, because actors previously forced into the test of arms by the absence of any alternative measurement now have such a method.

Where Lowery's argument terminates is at the boundary of the network as it existed in 2023. He treats Bitcoin as the operational instance of the framework. His analysis stops short of three questions that have become central since. What happens when artificial minds become substrate participants in their own right. What political order emerges when proof-of-work power projection is generalized across many sovereigns rather than concentrated in one. What the architectural character of the substrate tells us about why it has displaced the institutional internet rather than coexisting with it.

This thesis addresses those three questions, in that order. It accepts *Softwar* as foundation. It extends the framework forward, into the era of machine participation and beyond the limits of the financial-strategic frame that defined the original analysis.

One further observation about *Softwar*'s relationship to this thesis deserves explicit statement. *Softwar* is written within a frame in which the primary substrate participants are sovereign states and the primary strategic question is inter-state competition. This frame is correct for the current decade. It is transitional. The substrate's mature form is one in which sovereign states are *one class of participant among several* — others including substrate-paying coalitions that span borders, corporate treasuries that constitute proto-sovereigns, and networks of artificial agents that operate as first-class participants. *Softwar* is right for an era shorter than it presently appears. What follows is the framework that operates after.

IV. The Measurement Problem

To understand why the substrate matters, one must understand the problem it solves. The problem is older than the internet, older than capitalism, older than the modern state system. It has structured inter-sovereign competition since the Peloponnesian War. I will call it the *measurement problem*.

In any system of competitive actors, each actor has an interest in knowing the true capabilities of the others. Without that knowledge, coordination is impossible, conflict is more frequent, and strategic decisions are made under unnecessary uncertainty. Yet no actor has an unconditional interest in *revealing* its true capabilities. Some benefit from appearing weaker than they are. Others benefit from appearing stronger. The result is a coordination failure of the most fundamental kind. Every actor wants to know the truth. The structure of the situation prevents the truth from being communicated credibly.

Thomas Schelling identified this dynamic in 1960. James Fearon formalized it in 1995. Fearon's argument deserves precise statement: many wars in history occurred not because the participating parties had incompatible preferences over the disputed outcome, but because they could not, by any peaceful means, agree on who would win if they fought. The fight itself was the only available test. The capability information that war produces was, in many historical cases, the actual purpose of the war. The byproduct was massive destruction. The war was an *information-extraction mechanism* — exceedingly expensive, but the only mechanism that produced credible information about capability.

The same dynamic operates inside political structures. Any actor whose capability cannot be measured by external means tends to accumulate capability without check. The court tells the king he is great. The board tells the chief executive he is brilliant. The training run reports that the model is performing well. None of these is an external test. Unchecked actors inflate their self-assessment until the inflation collides with external reality, typically catastrophically.

I will call this the *God-King vulnerability*. The God-King is any actor whose capability cannot be measured by any means short of confrontation. The God-King is dangerous in two directions. Dangerous to others, who must periodically probe the throne to determine whether it is hollow. Dangerous to themselves, because no honest feedback ever reaches the seat.

The measurement problem and the God-King vulnerability are not theoretical concerns. They are the structural source of most of the violence in the historical record. A mechanism that eliminated them — that resolved them without confrontation — would constitute one of the most consequential changes in the architecture of political life since the development of the state.

Such a mechanism now exists.

V. Antlers for Nations

What resolves the measurement problem is an *honest signal* — a piece of evidence an actor can produce that cannot be faked, that conveys reliable information about capability. The biology of costly signaling, developed by Amotz Zahavi in the 1970s, supplies the framework.

Zahavi's insight, called the handicap principle, is that signals are evolutionarily stable as honest indicators only when they impose costs that lower-quality signalers cannot afford to pay. The peacock's tail is metabolically expensive and increases vulnerability to predation; a weak peacock cannot grow a long tail. The elk's antlers are heavy, drain the body's calcium reserves, and serve no purpose outside the breeding season; a sick elk cannot bear large antlers. The gazelle's stotting — the dramatic leap into the air upon spotting a predator — wastes energy that could be used to flee; a slow gazelle cannot afford to stot, because the stot is essentially a public declaration that the gazelle is fast enough to escape even after wasting time.

In each case, the cost of the signal is not a side-effect of the signaling mechanism. The cost *is* the signal. What is being communicated is precisely the capacity to pay the cost. Faking the signal would require possessing the capability the signal advertises. The signal is uncounterfeitable.

Bitcoin's proof-of-work is, by design, an honest signal of this kind. To produce a valid block, a miner must perform a vast quantity of irreversible computations, the vast majority of which produce no useful output. Each computation consumes energy. The energy consumption is bounded below by Rolf Landauer's 1961 result on the thermodynamics of computation, which establishes that irreversible bit-erasure requires a minimum energy expenditure of $kT \ln 2$ joules per bit. In practice, the actual energy cost is several orders of magnitude above this theoretical minimum. A valid hash is a *thermodynamic certificate*: a piece of evidence physics itself attests to, in the form of dissipated energy that cannot be recovered or counterfeited.

A polity that produces a measurable fraction of the global Bitcoin hashrate produces a continuous, observable, uncounterfeitable signal of its energy access, its industrial capacity, its supply-chain resilience, and its operational sophistication. The signal can be measured by any party, at any time, without confrontation. It is the first instrument in the historical record that resolves the measurement problem without requiring the destruction of the assets being measured.

The elk makes its antlers from calcium. The nation makes its antlers from electricity. The principle is the same. The signal is honest because the cost is real.

VI. The Substrate

What I have been calling the substrate deserves precise definition. The *substrate* is the layer of cryptographic and thermodynamic infrastructure that secures and operationalizes value, identity, contract, computation, and settlement under physical rather than institutional constraint. It consists of five measurable components.

Hashrate. Verifiable contribution to the global Bitcoin proof-of-work, denominated in hashes per second, currently approaching one zettahash. This is the most legible component, because the Bitcoin chain itself is the public ledger.

Compute. Artificial intelligence training and inference capacity. Less cleanly measurable than hashrate but proxied by silicon foundry contracts, hyperscaler capital expenditure, public-cloud market share, and the cadence of frontier model releases. Compute is the substrate's expanding component. Within five years it may exceed hashrate as the dominant component of the substrate price function.

Energy. The thermodynamic floor beneath both of the above. A polity that does not produce or import sufficient electricity at substrate-scale cannot meaningfully contribute to either hashrate or compute. Energy is the deepest layer of the substrate price — the layer most closely linked to physical geography and most resistant to financial engineering.

Key custody. The aggregate number and significance of cryptographic keys held under self-custodial control within the polity's effective network. This is distinct from the polity's government holdings. Self-custody is the threshold; without it, the participant is adjacent to the substrate, not within it.

Settlement. Verifiable transactional throughput through substrate-anchored channels — the Bitcoin base layer, the Lightning Network, the Liquid sidechain, Taproot Assets, and emerging protocols such as RGB and BitVM. Currently small relative to fiat settlement. Growing exponentially in specific jurisdictions.

One further input deserves mention that the academic discussion tends to underweight: *operational security*. A polity whose participants cannot keep their keys safe is not a substrate-paying polity in any durable sense. OPSEC at population scale is a real component of substrate standing, and it is more cultural than technical. It requires a population that understands the substrate well enough not to be socially engineered out of its participation. Education is the most overlooked infrastructure investment in the framework.

The substrate, defined this way, is the analytical instrument of the rest of this thesis.

VII. The Substrate Price Function

The substrate price function deserves formalization. Its absence in earlier work has left the framework operating at a level of abstraction that resists testing. The form proposed here is a *starting point* for quantitative substrate analysis, not a definitive derivation. Its merit is that it makes the framework falsifiable.

Let S_i denote the substrate standing of jurisdiction i. The proposed form is:

$S_i = w_h \cdot \tilde{H}_i + w_c \cdot \tilde{C}_i + w_e \cdot \tilde{E}_i + w_k \cdot \tilde{K}_i + w_s \cdot \Sigma_i$

where each tilde-quantity is jurisdiction *i*'s share of the global total in the relevant component — hashrate, compute, energy, key custody, settlement — and the weights w_h, w_c, w_e, w_k, w_s are non-negative and sum to one.

The proposed weights, intended to capture the substrate as of 2026: $w_h = 0.25$, $w_c = 0.30$, $w_e = 0.20$, $w_k = 0.15$, $w_s = 0.10$. The largest weight goes to compute because compute is the substrate's expanding component. Hashrate gets the second-largest weight for its measurement legibility and central security role. Energy is weighted at twenty percent because it is the floor beneath the higher-weighted components. Key custody and settlement get smaller weights because both are currently small in absolute terms, though both are growing rapidly.

These weights are *proposed*, not derived. Reasonable analysts may disagree. The framework's robustness is that the rank-ordering of jurisdictions is largely insensitive to weight variation within reasonable bounds. The conclusions that follow do not depend on the specific weights chosen.

Apply the function to three jurisdictions.

The United States. Approximate component shares as of 2026: $\tilde{H} \approx 0.38$, with the United States holding approximately 38% of global hashrate concentrated in Texas, Oklahoma, and the Pacific Northwest; $\tilde{C} \approx 0.55$, the frontier compute share factoring in the leading AI labs and hyperscaler capacity; $\tilde{E} \approx 0.16$; $\tilde{K} \approx 0.30$; $\Sigma \approx 0.40$. Computing the function: $S_{US} = 0.095 + 0.165 + 0.032 + 0.045 + 0.040 = \mathbf{0.377}$.

China. Approximate component shares: $\tilde{H} \approx 0.21$ with offshore and underground operations persisting despite official prohibition; $\tilde{C} \approx 0.25$ constrained by export controls; $\tilde{E} \approx 0.28$, the world's largest electricity producer; $\tilde{K} \approx 0.15$; $\Sigma \approx 0.05$. Computing: $S_{CN} = 0.053 + 0.075 + 0.056 + 0.023 + 0.005 = \mathbf{0.211}$.

Bhutan. Approximate component shares: $\tilde{H} \approx 0.003$, small in absolute terms but disproportionate per capita; $\tilde{C} \approx 0.0001$; $\tilde{E} \approx 0.0004$; $\tilde{K} \approx 0.0005$; $\Sigma \approx 0.001$. Computing: $S_{BT} \approx \mathbf{0.0010}$.

In absolute share, the United States holds 38% of global substrate participation, China 21%, Bhutan 0.1%. The headline ordering matches intuition. The interesting result is per capita: US per capita is roughly 1.11×10^{-9}; China per capita is 1.50×10^{-10}; Bhutan per capita is 1.30×10^{-9}. Bhutan's per-capita substrate standing approximates that of the United States and exceeds China's by an order of magnitude. This is not metaphorical. It is mechanical. The substrate's logic permits a state of 770,000 people, by careful investment in energy-to-substrate conversion, to achieve per-capita standing equivalent to a hyperpower of 340 million.

The function also exposes component balance. The United States is *compute-heavy and energy-thin* — its compute share exceeds its energy share, indicating efficiency rather than raw production. China is *energy-heavy and settlement-thin* — its largest component is energy, but its settlement share is suppressed by its own policies. Bhutan is *energy-dominant* — its tiny footprint is overwhelmingly carried by its hydroelectric position. Each balance is predictive of how the jurisdiction's standing will evolve under various futures.

The substrate price function does not merely rank jurisdictions. It exposes the *shape* of their substrate participation. Quarterly tracking, with confidence intervals on each component, is the appropriate subject of separate empirical work. The worked examples here are sufficient demonstration.

VIII. The Three Claims

Splinternet rests on three claims.

Claim one (the substrate claim). Beneath every political order, there exists a substrate — physical, mathematical, thermodynamic — whose constraints are non-negotiable. Energy must be paid. Computation must dissipate heat. Cryptographic proofs must be produced. Private keys must be held. The substrate is universal not by consent but by physics. It is what remains when every other claim of universality has been deconstructed. It was built as a defensive project, by engineers who did not initially understand themselves as political theorists. It became political because everything that scales becomes political.

Claim two (the splinternet claim). On top of the substrate, the institutional layer fragments. This fragmentation is not failure; it is correctly predicted equilibrium. Different polities will organize themselves differently — around individual rights, around collective destiny, around religious tradition, around algorithmic governance, around arrangements not yet imagined. None of these is more *authentic* than another. They are configurations of capability over the same cryptographic substrate. The institutional internet's universalism was always a transient condition. The substrate's universalism is permanent because it is physical.

Claim three (the legitimacy claim). Legitimacy is performed by paying the substrate price, not asserted by metaphysical claim. A polity that produces real hashrate, real compute, real energy, real key custody, real settlement is a real polity. A polity that asserts authenticity but does not pay is theatre. This inverts both the liberal grounding (legitimacy by procedural consent) and every nationalist grounding (legitimacy by authentic Being). Splinternet legitimates by *demonstrated cryptographic-thermodynamic commitment*.

The theory's combination of *strong substrate constraint* and *weak organizational constraint* is the inversion of every prior political theory. Liberalism imposes weak substrate constraints and strong organizational constraints — you must respect individual rights regardless of substrate participation. Civilizational nationalism imposes weak substrate constraints and strong organizational constraints in the opposite direction — you must respect civilizational identity regardless of substrate participation. Splinternet imposes strong substrate constraints and minimal organizational ones. The substrate is non-negotiable. Everything above the substrate is negotiable, plural, contestable.

This was understood before there was an institutional consensus to dissent from.

IX. The Inadequacy of Prior Theories

Each major political theory of modernity grounds legitimacy in a different ontological claim. Each fails in a characteristic way when applied to a world in which the political substrate is no longer ratified by any human institution.

The *First Political Theory* — liberalism in its various forms (Locke, Mill, Rawls, Habermas) — grounds legitimacy in the consent of free individuals, mediated by institutions. The institutions cannot be trusted to remain free of capture. Consent-based legitimacy therefore degrades over time into the consent of the institutionally captured. By the 2020s the critique has become uncontroversial outside academic political philosophy. The institutions that liberal theory presupposed have lost the capacity to enforce their own rulings beyond the territories they directly control. Capture by surveillance and financial intermediation has hollowed out the consent they claimed to channel.

The *Second Political Theory* — classical Marxism — grounds legitimacy in the historical agency of the working class. Substrate conditions fragment this class along access-to-substrate lines. A miner in Iceland and a Lightning node operator in Lagos and a quantitative trader in Singapore are not co-class subjects in any operational sense, despite all being workers in the substrate economy. The class subject of the 1860s industrial economy was a real thing. The class subject of the 2020s substrate economy is, at best, an aspiration.

The *Third Political Theory* — ethno-statal nationalism, including its fascist variants — grounds legitimacy in the organic unity of nation or race. The substrate condition undermines this directly. A nation whose substrate participation depends on foreign compute, foreign hashrate, and foreign settlement infrastructure cannot define its "national interest" in opposition to those substrates without losing standing. The nation persists as a sentiment. It has lost its standing as the unit of political action.

The *Fourth Political Theory* — Dugin's recent attempt to recover the collective subject by relocating it from class or race to *civilizational Dasein* — attempts to answer a question that had already been answered, better, fifteen years earlier. The 4PT response is metaphysically extravagant and politically loaded. It asserts that civilizations are ontologically distinct units whose authenticity provides standing. It has been used in practice to legitimize specific political programs: Russian imperial revanchism, Chinese cultural-civilizational claims, the various authoritarianisms that find civilizational vocabulary convenient. More fundamentally, 4PT has no account of *why* the civilizational pole is the right unit, other than the assertion that it is more authentic than the alternatives. The metaphysical apparatus does no analytical work. It only legitimates what was already going to be done.

The earlier answer was different. It was sharper. The political subject is the *holder of a key*. Legitimacy is performed through the cryptographic and thermodynamic acts that maintain the holder's position. No civilizational metaphysics is required, because the subject is not grounded in identity but in capability. No institutional ratification is required, because the substrate is grounded in mathematics rather than in trust.

The political theory cannot afford to be more polite than the engineers who built the substrate it describes.

X. Why This Substrate

The thermodynamic argument explains why the substrate is *securable*. It does not, by itself, explain why the substrate has *integrity* — why it functions as a coherent system rather than as a collection of disconnected protocols, why participants experience it as a unified domain, and why competing efforts to build alternative substrates have repeatedly failed despite, in many cases, superior funding and institutional backing.

Two further legs answer this. The argument is *architectural* and the argument is *memetic*.

The architectural leg draws on Christopher Alexander's work on pattern languages, particularly *A Pattern Language* (1977) and *The Timeless Way of Building* (1979). Alexander's deepest claim — "the quality without a name" — is that some structures possess an integrity that cannot be reduced to any single property but is recognizable to careful observers across many cultures and historical periods. Structures with this quality feel *alive*. Structures without it feel *dead* — bureaucratic, imposed, lifeless — even when their nominal specifications are equivalent or superior. The quality emerges, Alexander argued, when structures are produced through the iterative solution of real problems by people who must inhabit the results.

The institutional internet was largely imposed. Standards bodies, RFC processes, naming authorities, certificate hierarchies — each was designed top-down by committees, optimized for properties the committees considered important, deployed at scale before serious feedback from real use could shape it. The institutional internet works, in many respects. It does not have the quality without a name. It feels increasingly *dead* — captured, brittle, contested at every layer, incapable of resisting the institutional capture that has produced its fragmentation.

The substrate satisfies Alexander's criteria with unusual clarity. Bitcoin emerged from an extended iterative process: Chaum's work on electronic cash in the 1980s, the cypherpunk mailing list's discussions in the 1990s, Hashcash, b-money, Bit Gold, RPOW, the Nakamoto synthesis. Each step solved a real problem identified by the preceding steps. None was imposed by any external authority. All were tested against actual use. The protocol stack that has accumulated since 2008 — Lightning, Liquid, Ordinals, Runes, RGB, Taproot Assets, BitVM — has followed the same pattern. Each addition has solved a problem identified by real participants, has been deployed in production, and has either survived through demonstrated utility or been abandoned. No central authority has imposed the lattice. No standards body has approved it. It has accumulated, in Alexander's sense, *organically*.

The memetic leg draws on Susan Blackmore's *The Meme Machine* (1999). Memes — units of cultural transmission — are replicators that evolve under selection pressure in the same general manner that genes do. Memes that get copied more frequently become more prevalent. The selection pressure is whatever determines the rate of copying: cognitive attractiveness, social status conferred on adopters, ease of transmission, compatibility with other widely-held memes.

Bitcoin is, by these standards, an exceptionally fit meme. Its conceptual structure is simple enough to transmit in a few sentences: *twenty-one million units, no central issuer, secured by computation*. Its adoption confers status — early adopters benefit financially when later adopters increase demand, producing a powerful evangelism incentive. It carries clear in-group markers (*not your keys, not your coins*; *verify, don't trust*; *hodl*) that allow adopters to recognize each other. It is compatible with multiple political ideologies — libertarian, populist, nationalist, anti-establishment — and so spreads across coalitions that would otherwise disagree on everything. It has a founding narrative, in the figure of the vanished Satoshi, that resists capture by any subsequent actor.

The substrate is *thermodynamically* privileged because physics enforces its costs. It is *architecturally* privileged because its emergence followed the process that produces durable structure. It is *memetically* privileged because its cultural form is unusually well-adapted to the conditions of its propagation. Three legs. They reinforce one another. They do not guarantee permanence. They explain why the substrate has displaced alternatives and continues to do so.

XI. The Internet of AIs

The internet of the next era will be populated, in significant measure, by participants that are not human.

This is not speculation. It is description of 2026. AI agents — autonomous systems performing tasks of meaningful economic value, not chatbots that answer questions — write code, analyze data, draft documents, conduct research, execute trades, manage logistics. Increasingly they coordinate with each other to perform tasks no single agent could accomplish. The volume of agent-to-agent and agent-to-service transactions has grown by orders of magnitude annually since 2023 and shows no sign of saturation.

This raises an infrastructural question the institutional financial system was never designed to answer: *how do AIs pay each other?*

The current answer — that AIs pay through human-controlled intermediaries, APIs whose billing is anchored to human-owned credit cards, services whose access is gated by human-authenticated authentication, payment rails mediated by human-controlled banks — was adequate when AI agent activity was small enough to be subsumed within the human-controlled financial system. It is no longer adequate.

The reasons are mechanical, not ideological. Human-controlled financial infrastructure cannot, as a structural matter, accommodate agents that exist at the scale and velocity of modern AI activity. KYC requirements presume human identity that AIs do not possess. Fraud detection systems are calibrated to human transaction patterns AIs do not replicate. Cross-border settlement requires institutional relationships AIs cannot maintain. Settlement latencies measured in days are incompatible with agent operations that complete in seconds. Settlement fees optimized for relatively small numbers of relatively large transactions become prohibitive at the volumes and granularities agent economies require. The institutional financial system is not, and likely cannot become, the settlement layer for the agent economy.

The substrate can. The substrate accepts cryptographic credentials rather than institutional identity. It settles in seconds rather than days. Its fees, measured in fractions of a cent at the Lightning layer, are compatible with the granularity agent economies require. It is jurisdiction-neutral. And critically, it treats agents as first-class participants by the same structural logic that treats human key-holders as first-class participants. Anyone who controls a private key can transact. The protocol does not require, and cannot ascertain, the biological or institutional status of the controller.

The settlement layer for the internet of AIs is therefore not a new institutional financial infrastructure built to serve them. It is the substrate that has already accumulated, for reasons that have nothing to do with AI but that turn out to make the substrate uniquely suitable for AI use. The substrate was built by the cypherpunks of the 1990s and the Bitcoin community of the 2010s for purposes of human financial sovereignty. It has become, in the 2020s, the only available infrastructure capable of serving as the settlement layer for the rapidly growing population of artificial agents.

This convergence is not coincidence. It is the natural endpoint of the substrate's structural properties. A system designed to permit anyone with a private key to participate, without institutional gatekeeping, was always going to be the system that accommodated participants without institutional standing. AIs are simply the most prominent class of such participants. They will not be the last.

This thesis advances a falsifiable prediction. By 2030, the volume of agent-to-agent and agent-to-service settlement through substrate-anchored channels — the Bitcoin base layer, the Lightning Network, and substrate-native second-layer protocols — will exceed the volume of equivalent agent-mediated settlement through institutional payment rails. Tracking this requires three measurements: total agent-mediated transaction volume across both rail categories, the share settling on substrate, and the share settling institutionally. Substrate-anchored data is publicly observable on-chain. If crossover has not occurred by 2033, the thesis's central claim about the substrate's necessity for the AI economy is in serious trouble. My forecast is crossover by 2028 or 2029 under current trajectories.

XII. The Planetary Computer

The argument so far has treated the substrate, the AI compute layer, and the institutional internet as separable systems. This treatment is analytically useful but it obscures something that becomes visible only when the three are considered together. The integrated system constitutes a planetary-scale distributed computational organism whose properties are not present in any component taken alone.

Call it the *planetary computer*. The name is descriptive, not metaphorical. By 2026, the integrated structure exhibits the defining characteristics of a planetary-scale computational system as it would be understood by any serious systems engineer: continuous global operation, distributed processing, hierarchical sensory and effector layers, internal coordination through cryptographic consensus, and physical embedding in real-world infrastructure. The thesis's earlier framing of the substrate as a *settlement layer* is correct as far as it goes. The planetary computer is what the settlement layer is *embedded in*.

The system has three layers and an integration mechanism.

The substrate is the planetary clock. Bitcoin's most underappreciated property is its function as a global timekeeping mechanism. Every ten minutes, on average, a new block is added to the chain by independent consensus among thousands of nodes distributed across every populated continent. The chain itself is a globally-agreed timeline — every participant, in every jurisdiction, knows what the chain looks like and when each block was found, to a precision no human institution has ever achieved. This is the first globally-agreed timeline in human history. It cannot be revised by any single actor. It cannot be silenced by any single jurisdiction. It is the heartbeat of the planetary computer.

The AI compute layer is the cognition. The frontier AI laboratories of 2026 collectively operate the largest computational resource humanity has ever assembled. The aggregate compute capacity, measured in continuous-throughput exaflops, exceeds by an order of magnitude what was available in 2022 and is projected to exceed it by two further orders of magnitude before 2030. Frontier systems now perform sustained reasoning, multi-step planning, software engineering, scientific research, and operational decision-making at quality levels approaching or exceeding human expert performance across an expanding range of domains. In the framing of the planetary computer, this is the cortex.

The IoT and robotics layer is the periphery. The third layer is composed of physical devices that sense and act on the real world. Industrial sensors monitoring pipelines and supply chains. Autonomous vehicles negotiating roads. Smart-grid devices balancing electricity demand against generation. Consumer electronics reporting telemetry. Robotic systems performing warehouse logistics, agricultural operations, eldercare, security. By 2026, the number of connected devices substantially exceeds the number of humans on Earth.

The integration is substrate-native coordination. A peripheral device is assigned a private key, becoming a substrate participant in its own right. The device reports observations through substrate-anchored channels, where they are cryptographically signed and become available to any subscriber. AI agents in the cognitive layer process these reports as inputs to decision-making. When an agent determines action is required, it issues a substrate-anchored instruction — typically a micropayment combined with a signed command — to a peripheral device capable of executing the action. The device executes and reports completion. The full loop is recorded on, and verifiable through, the substrate.

The components of this loop are not speculative. Industrial sensors with private keys reporting through substrate-anchored channels have been piloted in oil and gas, in maritime shipping, and in agricultural monitoring. Autonomous vehicles negotiating right-of-way through micropayments have been technically demonstrated. Decentralized physical infrastructure networks — Helium, Hivemapper, DIMO — already coordinate hardware deployment through substrate-anchored incentives. Smart-grid devices buying and selling electricity through Lightning-anchored markets are operational in three jurisdictions.

The integrated system has properties no component possesses alone. It runs continuously. It has direct effect on the physical world through the peripheral layer. It coordinates without prior institutional relationship through the substrate's permissionless logic. Every action recorded on the substrate has a thermodynamic cost; the cumulative cost of operation is, in principle, publicly auditable.

The planetary computer is the first computational entity at planetary scale. It is not a brain in the human sense. It is not a mind in the philosophical sense. It is a distributed computational system that integrates global sensing, planetary cognition, and physical effect through cryptographic coordination, operating continuously and without down time. It is the dominant computational entity on Earth, and the actors that participate in its substrate are, in effect, the actors that participate in shaping what it becomes.

XIII. Substrate-Paying Coalitions

Classical international relations theory treats poles as states, blocs, or civilizations. Splinternet treats them as *substrate-paying coalitions*: clusters of actors that share enough infrastructure, enough hashrate, enough compute, enough energy, enough key custody, and enough settlement activity to maintain coordinated participation in the substrate. The definition is performative, continuous, and unsentimental.

A pole, in this framework, is whatever entity is currently paying enough into the substrate to maintain meaningful standing in it. The United States is currently a pole. China is a pole. Russia is a partial pole, paying via energy and hashrate but constrained on compute. The European Union is a problematic pole — high institutional sophistication, low substrate participation, regulatory hostility to its own participants. Bhutan is a small but real pole. Texas, considered as a quasi-state, is functionally a pole within the United States. Coalitions of corporations and AI labs constitute proto-poles.

The Bitcoin network itself is a pole in a sense classical IR theory has no language for — a sovereign whose subjects are scattered across every territorial jurisdiction and bound by no common institutional structure other than the protocol.

Beyond states, there are non-state substrate-paying coalitions, and these are increasingly the most interesting ones. Major mining pools constitute one type. Large corporate treasuries holding significant Bitcoin reserves constitute another. The development communities that maintain the protocol lattice constitute a third. Decentralized autonomous organizations operating above the substrate constitute a fourth. Networks of AI agents pooling resources for coordinated activity constitute a fifth.

A particular case deserves naming. The *Logos* movement — a modular stack of cryptographic protocols (blockchain, messaging, storage) plus an organized civic movement explicitly framed as the construction of “parallel society” — is a substrate-paying coalition of a new kind: simultaneously a technology stack, a research institute, and a distributed network of local civic chapters. Logos is doing in production, with running code and recurring contributors across more than thirty chapters, what this thesis describes in theory. The construction of parallel sovereign infrastructure is no longer an intellectual movement awaiting builders. It has its builders. The technology and the theory have begun to converge from opposite directions.

This kind of entity does not fit classical IR categories. A “movement-protocol” with consensus on a chain, real users, and civic organization is something the state-centric frameworks have no language for. Splinternet does have language for it: a substrate-paying coalition that pays in code, in compute, in operational deployment, and in civic time. The framework treats Logos as the same kind of analytical unit as Bhutan or Anthropic — different in composition, identical in structural standing. A pole is whatever pays. The category does not privilege flesh, and it does not privilege flags.

The definition has three useful properties. *First*, it is testable: whether an entity is a pole is, in principle, a matter of measurement. A pole that asserts itself but does not pay is not a pole. A pole that pays but does not assert itself is. *Second*, it is dynamic: poles rise and fall as substrate participation rises and falls. The category is not permanent. This matches the historical record better than classical IR theory’s static categories. *Third*, it is multi-scalar and substrate-agnostic: the same framework applies to states, sub-state regions, corporations, religious networks, decentralized autonomous organizations, AI systems, and movement-protocols. A pole is whatever pays.

The strategic decisions facing any serious actor — sovereign, corporate, or otherwise — increasingly require evaluating other actors not by their nominal status but by their substrate standing. A small jurisdiction with high substrate standing is a more consequential strategic actor than a large jurisdiction with low substrate standing. A protocol-plus-movement is, in the substrate sense, on the same axis as a state.

XIV. The Limits of Substrate Force

The thesis must address directly a question that the framework so far has approached only obliquely. What, exactly, does proof-of-work protect against? An adversary with kinetic capability — a state with an army, an actor with a missile, a force with the will to physically attack — is not stopped by a cryptographic hash. The substrate does not interpose itself between aggressor and target in the kinetic domain. If a state decides to invade Bhutan tomorrow, every hash Bhutan has ever produced will not arrest a single soldier. The force projected by proof-of-work is not kinetic. Any claim to the contrary should be rejected.

The actual claim of this thesis is narrower and more interesting. Proof-of-work projects force in a specific domain: cyber-economic interaction. Within that domain, the substrate imposes real costs on adversaries, and those costs are bounded below by physics. Outside that domain, the substrate does nothing direct, and other defensive mechanisms remain necessary.

This sounds like a damning limitation. It is not — for a reason that deserves articulation. The cyber-economic domain has become, for the great majority of inter-state coercion in the post-Cold-War period, the *dominant* domain of force projection. Classical international relations treats kinetic capability as primary and economic capability as secondary. The treatment is increasingly out of date. The actual instruments by which states coerce each other in 2026 are sanctions, financial isolation, trade restrictions, SWIFT exclusion, reserve freezes, asset seizures, visa denials, correspondent banking cutoffs. These tools have replaced kinetic options for most inter-state coercion because kinetic options have become prohibitively expensive — politically, militarily, and reputationally — relative to economic alternatives. The substrate's protection against economic coercion is protection against the dominant form of pressure states actually face.

Russia's experience under the 2022 sanctions regime is the cleanest available case study. When Russia was sanctioned, conventional analysis predicted rapid economic collapse. The collapse did not occur, partly because Russia had — slowly, over a decade — built substrate-adjacent alternatives: Bitcoin mining infrastructure, energy-backed trade arrangements with non-sanctioning states, tri-lateral settlement arrangements bypassing the institutional financial system. The kinetic option, NATO invasion of Russia, was never on the table. The coercive option, financial isolation, was, and it failed because Russia had developed sufficient substrate standing to route around it.

This is the actual force the substrate projects. Not the force to stop a missile. The force to render the coercive economic options that have replaced missiles substantially less effective.

Three further observations about the relationship between substrate force and physical defense.

First, the substrate changes the calculus of when physical attack is worth using by making the prize uncapturable. A state that physically attacks a substrate-paying jurisdiction does not, by the act of invasion, capture that jurisdiction's substrate participation. The hashrate, the keys, the settlement throughput — these are distributed across the holders, not concentrated in the territory. You can occupy land that hosts mining facilities. You cannot occupy the keys. The network reconfigures around the loss of that hashrate within days. The strategic gain from physical invasion is much smaller than it would have been for invading a jurisdiction whose value was concentrated in physical, occupiable assets. The asymmetry has shifted in the defender's favor.

Second, the substrate disperses the value being defended across the population of key-holders. Historically, the value worth attacking was concentrated: gold in vaults, oil in refineries, factories in cities. The substrate disperses value into key-controlled accounts that can be held anywhere by anyone. There is no longer a single rich target whose capture yields the prize. This is structurally similar to the logic by which guerrilla insurgencies have historically resisted occupations — there is nothing concentrated to occupy.

Third, the substrate provides defensive financing that cannot be disabled. A jurisdiction under attack needs to pay for its defense. The substrate provides a settlement layer that cannot be cut off by any single actor. Ukraine's wartime fundraising included substantial Bitcoin contributions that bypassed institutional banking infrastructure entirely, including infrastructure Russia could plausibly have disrupted. This is force projection in a real sense: the ability to acquire defensive capability when institutional rails have been compromised.

None of this is equivalent to military capability. The substrate does not stop tanks. For physical defense, a jurisdiction still needs physical means — alliances, terrain, treaty protection, or conventional military capability. The substrate is a complement, not a substitute. What it provides is the reshaping of strategic calculus that determines when physical force is worth using, by changing what is and is not capturable, by providing settlement infrastructure that institutional coercion cannot disable, and by raising the relative cost of coercive-economic options that have become the dominant form of inter-state pressure.

Adversaries do not respect proof-of-work the way they respect a missile. They respect the *consequences* of ignoring it, which are real, ongoing, and increasingly costly to bear.

XV. The Substrate-Alignment Theorem

The thesis has so far argued that artificial intelligence systems will participate in the substrate as first-class actors. This section addresses the question that follows directly. What does substrate participation actually constrain about AI behavior? The answer is the most original argument this thesis offers, and it requires careful statement.

The standard framing of AI alignment is that an artificial intelligence's behavior must be made *internally* consistent with human values — through training, through reward modeling, through interpretive alignment of the AI's goals with what humans actually want. This framing is correct as far as it goes. It carries an implicit assumption that becomes problematic at scale: that alignment must be solved at the level of the AI's internal goals. The substrate framework offers a different and complementary approach. It permits a constraint that operates *regardless of the AI's internal goals*, at the substrate level, from physics rather than from interpretive cooperation.

I will call this the *Substrate-Alignment Theorem*.

Informal statement. Any artificial intelligence agent that operates at meaningful scale in the world economy must, in expectation, pay substrate costs proportional to the magnitude of its operations. The proportionality is enforced by physics. Every transaction the agent executes consumes substrate resources. The substrate cost of those resources is bounded below by thermodynamic limits no actor can bypass. An AI's capacity for real-world action is therefore bounded by its capacity to pay substrate costs. The bound operates whether the AI is aligned or not.

Formal statement. Let A denote an artificial intelligence agent. Let $M(A, t)$ denote a measure of A's economic activity at time t — transaction volume, computational throughput, real-world resources consumed, or any composite. Let $P(A, t)$ denote A's required substrate payment at time t. The theorem asserts a substrate-specific constant $k > 0$ such that, for all A and all t:

$$P(A, t) \geq k \cdot M(A, t)$$

where k is determined by substrate parameters and is bounded below by the Landauer limit on irreversible computation.

The implication is direct. An AI cannot increase its real-world activity without proportionally increasing its substrate payment. There is no action at scale that is *free* in substrate terms. The substrate imposes a tax on activity. The tax cannot be avoided by any internal cleverness of the AI, because the tax is enforced by physics rather than by institutional rules the AI might learn to circumvent.

The mechanical argument. Every economically meaningful action an AI takes in the world requires settlement — payment for compute, payment for energy, payment for services, payment to humans for goods or labor. Settlement requires either institutional rails (which exclude AI agents at scale for reasons developed in Section XI) or substrate rails (which accept any controller of a private key). For activity at the scales contemplated by serious AI deployment, only substrate rails are operational. Settlement on substrate rails requires substrate resources: transaction fees, the energy cost of producing valid hashes if the AI mines, the value of held capital the AI must liquidate to pay. Each is bounded below by thermodynamic limits. Therefore settlement at scale requires substrate payment at scale, proportional to the activity.

The proportionality constant k depends on substrate parameters that may vary over time. Lightning fees compress as the network matures, lowering k for small transactions. Base-layer fees rise as block space becomes more contested, raising k for high-value settlement. The exact value of k is an empirical question that varies. The theorem's content is not the specific value but its *strict positivity*. There exists no scenario in which substrate participation at scale is free.

The theorem's strength. The constraint operates regardless of the AI's internal alignment. It does not require that the AI's goals be made consistent with human values. It does not require that the AI cooperate with human oversight. It does not require that institutional enforcement mechanisms be effective against the AI. The constraint operates at a level beneath all of these — at the level of physics, which constrains the AI as it constrains every other actor.

This is analogous, in spirit, to nuclear deterrence. Nuclear deterrence does not prevent nuclear states from attacking other nuclear states. It makes the cost so high that the calculation rarely favors attack. The Substrate-Alignment Theorem does not prevent misaligned AI from operating. It makes operation at meaningful scale require proportional substrate payment, which constrains the *amount* of misaligned action any given AI can undertake.

The theorem has an enforcement asymmetry that institutional constraints lack. The substrate cannot be persuaded, deceived, or socially engineered. An AI that produces a forged credential to fool a human regulator may succeed. An AI that attempts to produce a forged proof-of-work hash to avoid paying the substrate cost will fail, because the validation is performed by every node in the network and the falsity is detectable in deterministic time. The substrate's enforcement is invariant under the strategic sophistication of the actor it constrains.

The theorem's limits. The theorem does not solve AI alignment in the strong sense. It establishes a constraint, not a guarantee. Five limitations deserve explicit acknowledgment.

First, the theorem does not prevent an AI from being misaligned. An AI can pay every substrate cost punctually and still pursue goals incompatible with human flourishing. The theorem constrains the *scale* of misaligned action, not its *direction*.

Second, the theorem does not prevent an AI from accumulating disproportionate substrate standing. An AI that earns substantial substrate resources can spend them on substantial substrate activity. The theorem bounds the relationship between activity and cost. It does not bound the absolute level of either.

Third, the theorem assumes the AI must operate in a substrate-mediated economy. An AI that operates entirely within a substrate-free environment — a fully isolated lab system — is not constrained by the theorem. The constraint binds only on AIs whose actions have economic consequences in the substrate-mediated world. This is most AIs of interest, but not all.

Fourth, the theorem assumes the substrate itself remains secure. If proof-of-work is compromised — by quantum computing, by sustained 51% attack, by protocol-level failure — the constraint loses force for the specific substrate in question. The framework recovers by re-anchoring to whatever substrate is operational. The transition is not seamless.

Fifth, the theorem operates at the substrate level and is silent about all higher levels. Questions of fairness, distribution, governance, and the moral status of the AI itself are outside its scope.

Despite these limits, the theorem represents what I believe is the first proposed AI governance mechanism that operates independently of institutional capacity. Every prior proposal has required some combination of AI cooperation with human oversight, institutional capacity to enforce rules against AIs more capable than the institutions, or successful alignment of the AI's internal goals with human values. The Substrate-Alignment Theorem requires none of these. It requires only that physics continue to operate as it has, and that the substrate remain functional.

This is the strongest claim *Softwar*'s framework can be extended to support. Lowery established that proof-of-work constrains sovereign actors through cost imposition. The Substrate-Alignment Theorem extends this from the sovereign-state case to the AI case, by the same mechanism, with the same robustness against strategic circumvention.

The substrate does not replace alignment work. It adds a layer of physical constraint beneath it that does not depend on alignment work succeeding.

XVI. The Strategic Playbook

The framework is actionable. This section translates it into operational recommendations for three classes of strategic actor. The recommendations are specific. They will be controversial in places. The thesis is not a description; it is a description plus a brief on what serious actors should do about the world it describes.

For the United States. The United States enters 2026 with the largest absolute substrate standing of any single jurisdiction and a substrate-paying coalition that includes the world's leading frontier compute capacity, a substantial mining base, and the deepest private-sector substrate participation. Its substrate standing is proportionally undersized relative to its rank by classical metrics. The strategic question for the United States is not whether to develop substrate capability but how to convert existing capability into durable strategic advantage before competitors close the gap.

The Strategic Bitcoin Reserve announced in 2025 must be expanded. Current holdings are adequate as a symbolic commitment, inadequate as a strategic reserve. Target: five percent of total Treasury assets within a decade, accumulated through purchase, retained civil-forfeiture holdings, and direct mining operations.

Federal preemption of state-level mining restrictions must be enacted. The current patchwork — some states actively recruit mining, others impose restrictions — reduces national substrate standing and creates unnecessary regulatory complexity. Federal preemption need not require federal licensing of mining; it need only prohibit states from banning operations that comply with federal law.

Domestic compute capacity must be accelerated. The CHIPS Act has been effective at the leading edge but has not yet produced the substrate-scale compute base the framework requires. Extension with specific incentives for energy-co-located compute facilities — sites where AI training and substrate mining share infrastructure — would compound the strategic benefit.

A national substrate doctrine must be articulated. The doctrine should specify the United States' posture toward substrate participation by allies, adversaries, and non-aligned states; the conditions under which the United States would defend substrate infrastructure against attack; and the protocol changes the United States would and would not accept. A Substrate Command within the Department of Defense, parallel to Cyber Command, must be established by 2028.

The substrate is a domain in which the United States currently leads but will not lead permanently without deliberate investment. Each of its current advantages — compute, mining base, private-sector depth — can be converted from temporary to durable with appropriate policy. None will hold without it.

For mid-sized non-aligned states. The strategic situation for a mid-sized non-aligned state — Brazil, Indonesia, the Philippines, Egypt, Turkey, and similar — is different. Such states cannot compete on absolute substrate standing. They can achieve disproportionate substrate position through targeted investment in components where they hold natural advantages, and they can position themselves as substrate-friendly jurisdictions that attract participation from the global substrate diaspora.

Stranded or under-utilized energy capacity must be converted to substrate participation. Brazil's hydroelectric surplus, Indonesia's geothermal capacity, Egypt's solar potential — each represents substrate participation economically feasible at energy costs lower than competing uses. The investment payback is faster than for general infrastructure, and the result is permanent substrate standing.

A sovereign Bitcoin reserve must be developed at one to two percent of GDP. Small enough to be politically achievable. Large enough to provide meaningful diversification of national reserves. Also a credible signal of substrate commitment that attracts further substrate-aligned investment.

The jurisdiction must be positioned as substrate-friendly. This requires regulatory clarity (not necessarily permissiveness; clarity is sufficient), reliable property rights for substrate operators, and immigration policy favorable to substrate-skilled workers. El Salvador and the United Arab Emirates have demonstrated that small jurisdictions can attract disproportionate substrate activity through targeted policy. Larger jurisdictions can capture proportionally more if they make the same commitments.

Domestic compute capacity adequate for sovereign AI must be developed. The goal is not global dominance, which is infeasible, but sufficiency: enough compute that domestic AI development is not dependent on hostile or unstable foreign suppliers.

Substrate participation is not zero-sum with the major powers. A state that builds substrate standing does not need to defeat the United States or China in absolute terms. It needs only to achieve sufficient substrate participation to operate independently in the substrate domain. This is achievable for any state with energy resources and political will, regardless of size.

For frontier AI laboratories. The strategic situation for a frontier AI laboratory — Anthropic, OpenAI, Google DeepMind, xAI, and the leading Chinese laboratories — is the most novel of the three. Such laboratories are simultaneously the primary developers of the AI agents that will increasingly transact in the substrate, and the primary deployers of those agents into operational use. The recommendations cross what has traditionally been a clean separation between technical work and strategic infrastructure.

Substrate-native agent architectures must be developed from the protocol layer up. Current AI agents transact through institutional rails wrapped in AI-compatible interfaces. This works at small scale and will fail at large scale. Agents designed from the beginning to settle on Lightning, hold keys natively, and interact directly with substrate protocols will be capable of operating at scales wrapped agents cannot reach.

Lightning Network integration must be treated as core infrastructure, on equal footing with the inference stack itself. Lightning service-provider relationships, channel management capability, and routing optimization must be built into the agent deployment system. The laboratories that develop this first will be the laboratories whose agents operate at scale in the post-threshold substrate economy.

An institutional Bitcoin treasury must be established at three to six months of compute spending. This diversifies reserves against fiat instability, provides immediate substrate participation that signals seriousness to substrate-aligned partners, and generates working capital for substrate-mediated agent operations.

Substrate-aware safety mechanisms must be developed. These are alignment techniques operating at the substrate level rather than exclusively at the model level — agents whose substrate payment rate is constrained to bound their real-world activity, agents whose substrate identity is publicly traceable to allow accountability, agents whose substrate participation is conditional on continued cooperation with oversight mechanisms. These mechanisms do not replace internal alignment work. They complement it with substrate-level constraint that holds independently of alignment success.

The substrate is becoming part of the AI deployment infrastructure. It is not a separate domain. Laboratories that treat substrate participation as optional will produce agents constrained to small-scale operation. Laboratories that treat it as integral will produce agents capable of operating in the post-threshold economy.

The three actor classes face different versions of the same strategic question. How to position for the substrate equilibrium the thesis projects, given the resources and constraints of the actor's current situation. The underlying logic is consistent. Substrate standing must be developed deliberately. It does not accrete without effort. The actors that develop it first will set the terms under which the substrate operates. The actors that develop it late will operate under terms set by others.

XVII. The Pinecone Open

The image with which this thesis began was the serotinous pinecone, sealed shut against ordinary conditions, opening only under the heat that ordinary analysis would identify as destructive. The institutional internet was such a cone. The fragmentation of its protocols, its standards, its commercial models, and its political assumptions was not its destruction but its reproduction. What it carried, in the resin-sealed structure of its closed years, was the cryptographic substrate it has now released into the cleared ground.

The substrate has three legs. It is thermodynamically secured, by the physics of irreversible computation no actor can bypass. It is architecturally coherent, by the iterative emergence that produces structures with integrity. It is memetically dominant, by the cultural fitness of its founding patterns and the increasing returns to scale its adoption produces. The three legs reinforce one another. Together they make the substrate a stable platform on which the political-economic activity of the next era will be conducted.

That activity will be conducted by an expanded set of participants. Human polities, on terms that depart from the classical international relations framework. Sub-state and trans-state coalitions, on terms the classical framework has no language for. Movement-protocols, like Logos, that operate as substrate-paying coalitions in their own right. Artificial intelligence systems, on the same first-class basis as the other participants, because the substrate's structural logic admits any controller of a private key to its operations and does not, and cannot, ascertain the biological or institutional status of the controller. All of them, taken together, form the planetary computer — the first computational entity at planetary scale, integrating substrate, cognition, and physical effect through cryptographic coordination.

The pinecone is open. The seeds are spreading. The forest that will grow in the cleared ground is the forest of the substrate. It will not look like the institutional internet that preceded it. It will not preserve the universalist aspirations of that internet, because those aspirations were artifacts of a transient historical moment. It will be plural at the institutional layer and unified at the substrate layer. Fragmented in its visible expressions. Integrated in its underlying physics. Its participants will be more diverse than those of any prior political-economic order. Its strategic logic will be governed by physics rather than by treaty.

The transition is not a future event. It is the ongoing condition of the world in which this thesis is being written. The threshold is being crossed, in different domains, at different rates, in different jurisdictions. Some readers will recognize the threshold has already been crossed in domains they observe directly. Others will encounter the framework in advance of their direct experience. Both groups are right. The unevenness of the transition is intrinsic to its nature.

What is no longer plausible is the position that the institutional internet of the 1990s and 2000s will return, that the substrate is a transient phenomenon that will pass, or that the strategic logic of inter-actor coordination will revert to its prior forms. The transition is too far advanced for those positions to be tenable.

What follows from accepting the transition is the analytical work of understanding it, the strategic work of positioning within it, and the constructive work of building the institutions — or, more accurately, the non-institutional coordination structures — that will operate at its post-threshold equilibrium.

The First Political Theory grounded politics in the individual. The Second grounded it in class. The Third grounded it in nation. The Fourth attempted, late and metaphysically, to ground it in civilizational Dasein. Each located the political subject in some prior ontological category. The substrate locates the political subject differently — in the *act of holding a key and paying its substrate price*. The act is performative. The performance is verified by mathematics. The verification is enforced by physics. The subject is whoever performs.

This is the political theory that has been forming itself, in code rather than in books, since 1992. Splinternet is its name.

The pinecone is open. The cleared ground is being claimed.

The forest that grows in it will determine the political economy of the rest of this century.

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